

1. U attributes

@d_U

Same ordered group `J_sel`

Y, P_Y
positions(P_Y, J_sel)
→ `lhs_contract`

Z, P_Z
positions(P_Z, J_sel)
→ `rhs_contract`

2. Establish P_U

```
f_Y = ordered_complement(
    ρ(P_Y), @d_U.lhs_contract)

f_Z = ordered_complement(
    ρ(P_Z), @d_U.rhs_contract)
```

```
P_U = gather(P_Y, f_Y)
      ++ gather(P_Z, f_Z)
```

Extend axis mappings to U

Preserved groups

`K` `J_rem` `I_sel` `F`

3. Final attributes

@d_final

Same ordered group `G = I_sel ++ K`

X, P_X
positions(P_X, G)
→ `lhs_contract`

U, P_U
positions(P_U, G)
→ `rhs_contract`

Constructed contraction attributes

Attribute	@d_U	@d_final
lhs_contract	positions(P_Y, J_sel)	positions(P_X, G)
rhs_contract	positions(P_Z, J_sel)	positions(P_U, G)

positions(P, G) returns p with gather(P, p) = G.
ρ(P) = range(|P|); ++ concatenates sequences.